ECE 105: Introduction to Electrical Engineering

Lecture 7

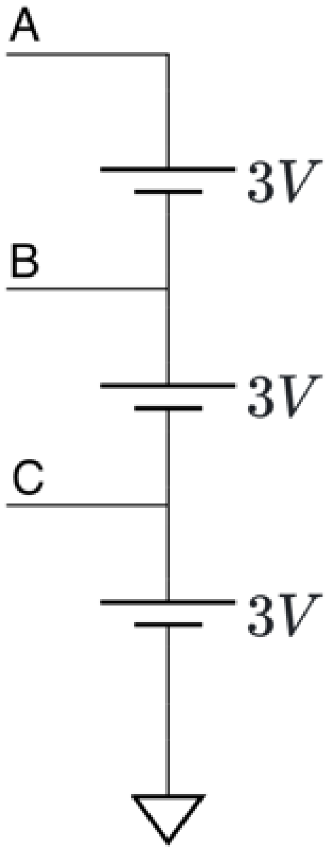
Circuit Problems

Yasser Khan

Rehan Kapadia

Problem 1

Given the circuit diagram with nodes **A, B,** and **C**, each connected to **3V** sources as shown. Determine the voltages $\mathbf{V_A}$, $\mathbf{V_B}$, and $\mathbf{V_C}$ at the respective nodes with respect to the ground.



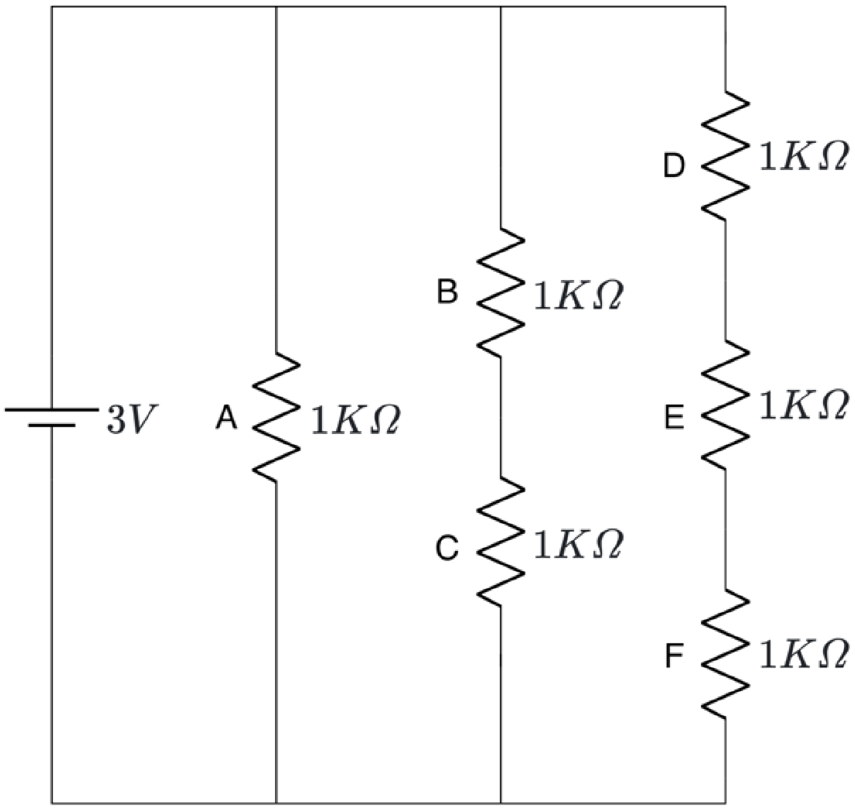
KCL

Voltage divider

Current divider

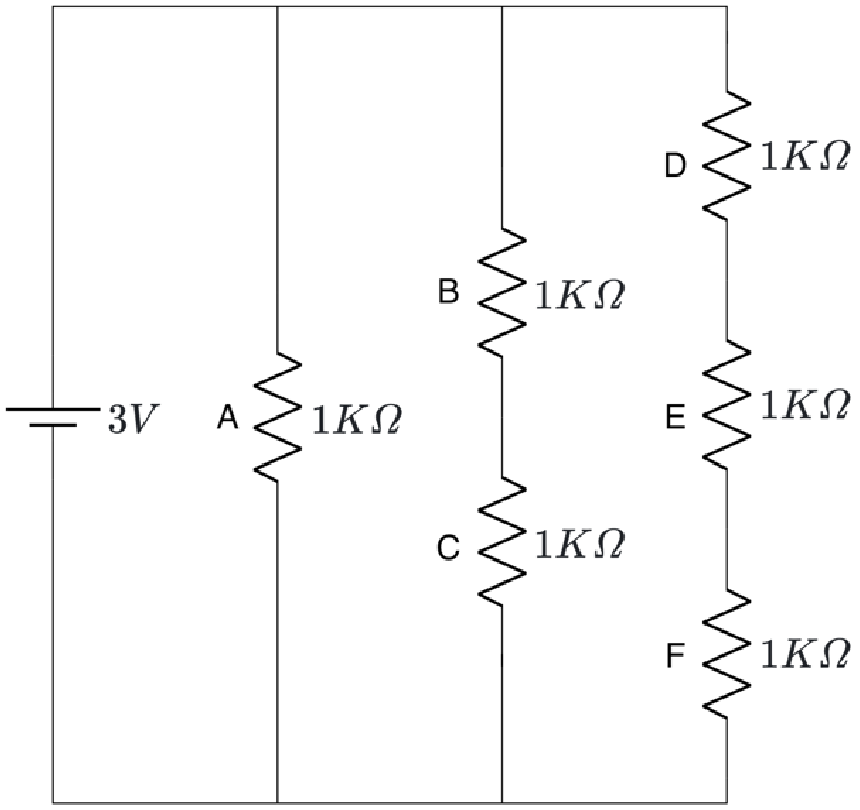
Problem 2

Given the circuit with six resistors connected to a **3V** voltage source as shown. Find voltages across the resistors (**V_A, V_B, V_C, V_D, V_E,** and **V_F**) and current through resistors (**I_A, I_B, I_C, I_D, I_E,** and **I_F**).



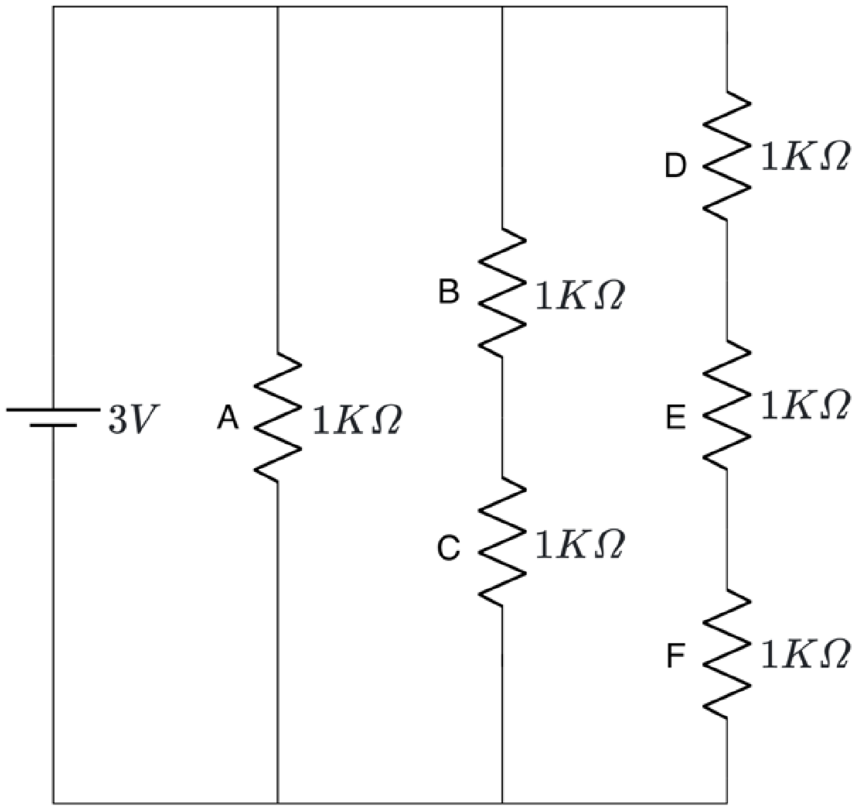
Problem 2

Given the circuit with six resistors connected to a **3V** voltage source as shown. Find voltages across the resistors (**V_A, V_B, V_C, V_D, V_E,** and **V_F**) and current through resistors (**I_A, I_B, I_C, I_D, I_E,** and **I_F**).



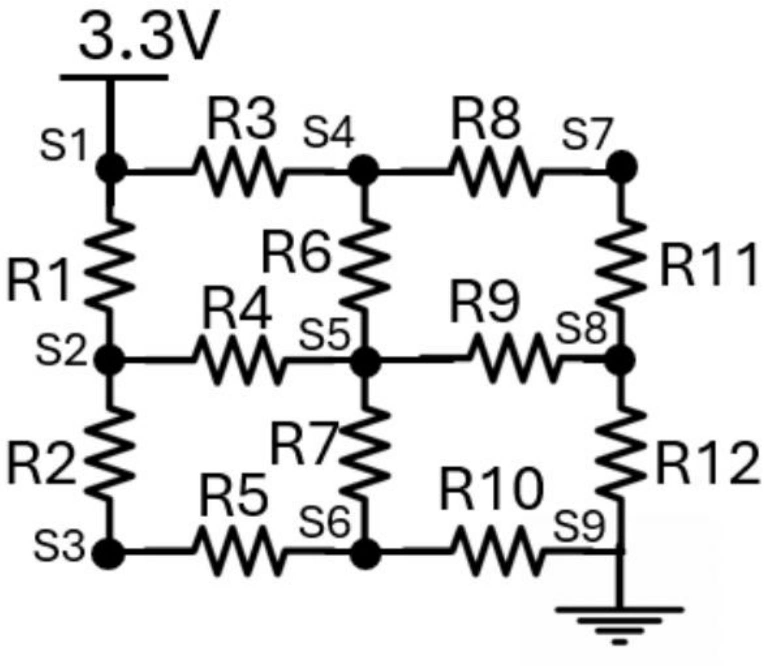
Problem 2

Given the circuit with six resistors connected to a **3V** voltage source as shown. Find voltages across the resistors ($\mathbf{V_A}$, $\mathbf{V_B}$, $\mathbf{V_C}$, $\mathbf{V_D}$, $\mathbf{V_E}$, and $\mathbf{V_F}$) and current through resistors ($\mathbf{I_A}$, $\mathbf{I_B}$, $\mathbf{I_C}$, $\mathbf{I_D}$, $\mathbf{I_E}$, and $\mathbf{I_F}$).



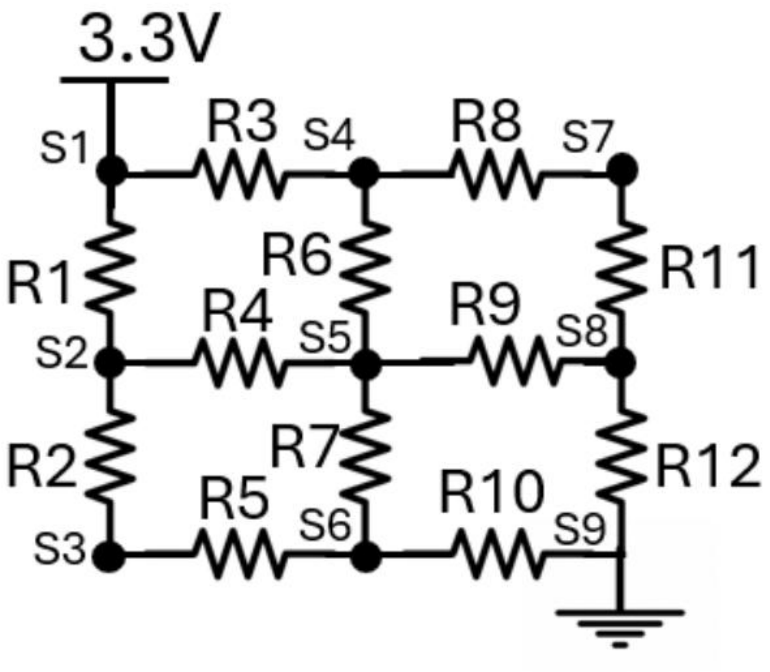
Problem 3

Given the circuit diagram with nodes **S_1 to S_9**, **S_1** is connected to **3.3V** and GND (0V). Determine the voltages **S_1 to S_9** at the respective nodes with respect to the ground if **R_1 to R_{12}** = R



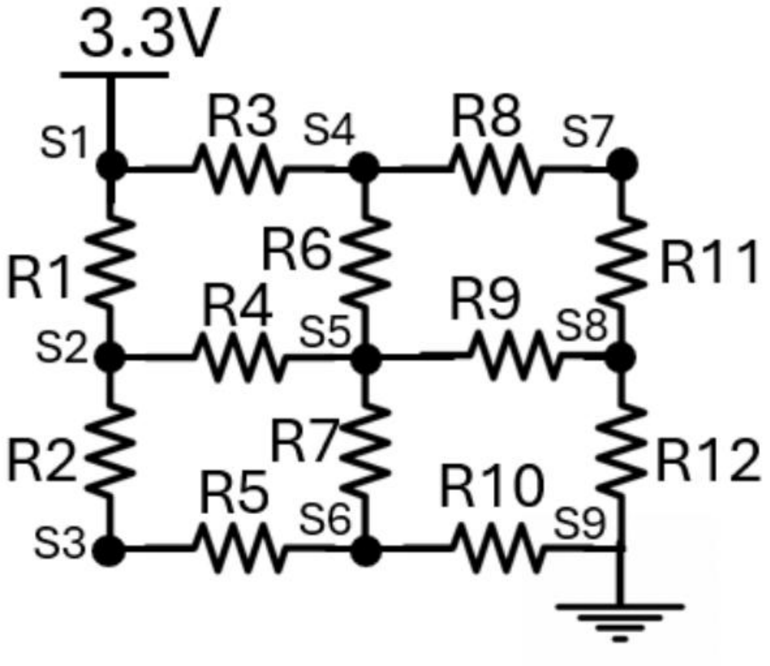
Problem 3

Given the circuit diagram with nodes **S_1 to S_9**, **S_1** is connected to **3.3V** and GND (0V). Determine the voltages **S_1 to S_9** at the respective nodes with respect to the ground if **R_1 to R_{12}** = R



Problem 3

Given the circuit diagram with nodes **S_1 to S_9**, **S_1** is connected to **3.3V** and GND (0V). Determine the voltages **S_1 to S_9** at the respective nodes with respect to the ground if **R_1 to R_{12}** = R



Python notebook